Rufus is a popular, free, and open-source utility for Microsoft Windows that formats and creates **bootable USB flash drives**. While Ventoy is better for multi-booting many systems from one drive, Rufus is known for being **faster** and providing deeper **customisation** for single-OS installations.

## Key Features

- **Speed:** Frequently outperforms similar tools, creating bootable media up to twice as fast as competitors.
- **Windows 11 Bypasses:** It can automatically remove requirements for **TPM 2.0, Secure Boot**, and mandatory Microsoft accounts during the setup process.
- **Direct Downloads:** You can download official retail ISOs for Windows 8.1, 10, and 11 directly through the Rufus interface.
- **Portability:** The software is a "portable" app, meaning it runs as a single `.exe` file with no installation required.
- **Broad Compatibility:** Supports both modern **UEFI (GPT)** and older **Legacy BIOS (MBR)** systems.

## Official Download Link

You can download the latest version (v4.14 as of late April 2026) directly from the developer:

- **Official Website:** rufus.ie
- **GitHub Releases:** pbatard/rufus GitHub

## Rufus vs. Ventoy: Quick Comparison

| Feature | Rufus | Ventoy |
|---|---|---|
| **Best For** | Fast, single-OS installs with custom tweaks | Multi-booting many ISOs from one drive |
| **Usage** | Formats and writes one ISO at a time | One-time install; just drag-and-drop ISOs |
| **OS Support** | Windows Only | Windows, Linux, and macOS |
| **Persistence** | Basic support for some Linux distros | Advanced persistence support |

Rufus allows you to bypass Windows 11 hardware requirements by automatically injecting a script into the installation media that skips compatibility checks.

## Step-by-Step: Bypass TPM, Secure Boot, & CPU Checks [4]

1. **Get Your Files:** Download the latest [Windows 11 ISO]() from Microsoft and the [Rufus executable]().
2. **Initial Setup:**
   - Open **Rufus** and insert a USB drive (8GB or larger).
   - Select your USB under **Device**.
   - Click **SELECT** and choose your Windows 11 ISO.
3. **Configure Partitioning:**
   - Set **Partition scheme** to **GPT** for modern UEFI systems or **MBR** for very old BIOS systems.
   - Set **Target system** to **UEFI (non-CSM)**.
4. **The Bypass Menu:** Click **START**. A "Windows User Experience" window will pop up. This is where you select your bypass options:
   - **Remove requirement for 4GB+ RAM, Secure Boot and TPM 2.0:** Check this to install on unsupported hardware.
   - **Remove requirement for an online Microsoft account:** Check this if you want to set up a local account instead of signing in.
   - **Create a local account with username:** Check this and type a name to skip the initial setup questions.
5. **Create the Drive:** Click **OK**. A final warning will tell you the USB will be wiped. Click **OK** again.
6. Installation: Once Rufus says "Ready," plug the USB into your unsupported PC and boot from it. The installer will no longer block you with the "This PC can't run Windows 11" message.

### Important Limitations

- **CPU Hard Blocks:** For Windows 11 24H2 and newer, your CPU **must** support the **SSE 4.2** instruction set. This is a hardware requirement that Rufus cannot bypass; if your CPU is too old (e.g., Intel Core 2 Duo), the OS simply won't boot.
- **Microsoft Support:** Systems running Windows 11 on unsupported hardware may not receive all security updates or future feature patches.

Identifying if your CPU supports **SSE 4.2** is critical for Windows 11 (version 24H2 and newer), as this is a physical hardware requirement that cannot be bypassed with tools like Rufus.

## Method 1: Using CPU-Z (Windows)

This is the easiest visual method for Windows users. [3]

1. Download and install the CPU-Z utility from CPUID.
2. Launch the application and stay on the **"CPU"** tab.
3. Look at the **"Instructions"** field.
4. Check for **"SSE4.2"** in the list. If it's there, your processor is compatible with the latest Windows 11 updates.

## Method 2: Using the Command Line

If you prefer not to install new software, you can use these built-in commands:

- **For Windows (PowerShell):**
  Open PowerShell as an administrator and run this command to check for the specific feature:
- `(Get-WmiObject Win32_Processor).Caption`

  *Note: While this gives the CPU name, a more reliable technical check in PowerShell uses a specialized script to call the* `IsProcessorFeaturePresent` *function with code 38.*

- **For Linux:**
  Open a terminal and run:
- `grep -o 'sse4_2' /proc/cpuinfo`

  If it returns `sse4_2`, your CPU supports it. If it returns nothing, it does not.

## Method 3: Official Specifications

If you know your processor model (e.g., Intel Core i5-2500K), you can look it up directly:

- **Intel:** Search for your model on the Intel Product Specification (ARK) page under "Advanced Technologies".
- **AMD:** Check the "Supported Extensions" section on the official AMD product page.

**Quick Tip:** Most Intel CPUs from the **Nehalem architecture** (approx. 2008) and AMD CPUs from the **Bulldozer/Ryzen** eras (approx. 2011) and newer typically support SSE 4.2.